

Tutorial Preparation Materials

26th European Agent Systems Summer School
Malmö University · 21–23 September 2026

Purpose of this document. This document consolidates the preparation materials supplied by the EASSS 2026 tutors. Please follow the tutor-specific distinction between prerequisite, recommended, and optional material. Where a tutor has provided files separately, those files are not reproduced here.

1. From Micro-Behaviors to Macro-Phenomena: An Introduction to Agent-Based Modeling

Prerequisite readings

- [Joshua M. Epstein \(2008\), JASSS, “Why Model?”](#)
- [Edmonds et al. \(2019\), JASSS, “Different Modelling Purposes”](#)

Further reading

- Dignum, F. (Ed.). (2021). *Social Simulation for a Crisis: Results and Lessons from Simulating the COVID-19 Crisis*. Springer.

This reading is recommended for participants interested in how social simulation can inform crisis response, support policy discussions, and reveal the potential consequences of alternative interventions.

Software and preparation

- Install the latest version of NetLogo. [NetLogo website](#) → Products / NetLogo Desktop → Download.
- If NetLogo is not installed locally, participants can use the web interface for the exercises, but local installation is recommended.
- After a theory background lecture, the tutorial will proceed to coding exercises in NetLogo, including learning syntax, debugging code, and building a basic model.

2. Formal Methods for Safe Multi-Agent Reinforcement Learning

Software used in the demo

The tutors use the MASA-Safe-RL Python package for interactive parts of the tutorial.

- Python with Jupyter Notebook installed.
- A Python setup can be installed with VS Code, required dependencies, and Jupyter Notebook extensions for VS Code. The tutor suggests [uv](#) as an easy option.
- Download/clone the [MASA-Safe-RL GitHub repository](#).
- If installation is successful, try running Notebook 1 under the tutorial/ folder and check that the output is correct without package errors.

Further readings

- Chapters 1 and 2 of Michael Huth & Mark Ryan, Logic in Computer Science - Modelling and Reasoning about Systems (2nd ed.). Cambridge University Press, 2004.
- Chapter 5 of Principles of Model Checking. MIT Press, 2008.
- Parts 1 and 2 of Richard S. Sutton & Andrew G. Barto, Reinforcement Learning - An Introduction (2nd ed.). MIT Press, 2018.
- Parts 1 and 2 of Stefano V. Albrecht, Filippas Christianos & Lukas Schäfer, Multi-Agent Reinforcement Learning: Foundations and Modern Approaches. MIT Press, 2024.
- D. Amodei et al., Concrete Problems in AI Safety, 2016.
- Bengio et al., Superintelligent Agents Pose Catastrophic Risks: Can Scientist AI Offer a Safer Path?, 2025.
- Mohammadhosein Hasanbeig, Daniel Kroening & Alessandro Abate, Deep Reinforcement Learning with Temporal Logics. FORMATS 2020.
- Mohammed Alshiekh, Roderick Bloem, Rüdiger Ehlers, Bettina Könighofer, Scott Niekum & Ufuk Topcu, Safe Reinforcement Learning via Shielding. AAAI 2018.
- Ingy Elsayed-Aly, Suda Bharadwaj, Christopher Amato, Rüdiger Ehlers, Ufuk Topcu & Lu Feng, Safe Multi-Agent Reinforcement Learning via Shielding. AAMAS 2021.

Tutor note: These readings are recommended, not required. There are no prerequisites and no knowledge of reinforcement learning is assumed, but understanding the basics of propositional and first-order logic would be useful.

3. Multi-Agent Reinforcement Learning for Traffic Control

Further preparation

1. Read Chapters 1 and 2 of the book provided separately by the tutor. **Important: students are asked NOT to share or make the book available in any form, as this may cause problems for the authors with the publisher. To get access to the book, please write an email to easss2026@mau.se**
2. Read the recommended article: [Google Scholar citation / PDF](#)
3. Read the arXiv paper provided separately by the tutor and complete the [SUMO Hello World tutorial](#).

SUMO note: It is not mandatory for students to come with SUMO installed. Ana will demonstrate the basic steps on her computer. However, participants are strongly encouraged to try the tutorial beforehand, as SUMO may not work on all operating systems. Ana uses Linux.

Optional readings

- Chapter 3 of the book provided separately by the tutor.
- If interested in reinforcement learning: [optional RL reading](#)
- If interested in agent-based modelling and simulation: [optional ABM/simulation reading](#)
- If interested in a broad overview of uses of AI in transportation: [optional AI in transportation reading](#)

4. Multiagent AI

Software

- Install [BSPL](#).
- Consider installing [Langshaw](#) (the tutor notes that it is still under development).

Further Reading List

- [arXiv 2606.05043](#)
- [arXiv 2606.05390](#)
- [Computer ACL paper](#)
- [AI + Law paper](#)
- [AAMAS 2011 - IBIOP](#)
- [AAMAS 2012 - BSPL](#)
- [TIST 2013 - Governance](#)
- [SCC 2014 - Bliss](#)
- [AAAI 2015 - Cupid](#)
- [AAMAS 2016 - Custard](#)
- [IJCAI 2021 - Tango](#)
- [AAMAS 2023 - Kiko](#)
- [IJCAI 2024 - Langshaw](#)
- [IJCAI 2025 - Mambo](#)
- [AAAI 2025 - Orpheus](#)
- [AAMAS 2025 - Azorus](#)
- [AAMAS 2026 - Peach](#)

Certificate and ECTS

EASSS 2026 recommendation: 3 ECTS

The estimated workload for this summer school is 75 hours, comprising 25 hours of in-person instruction and 50 hours of preparatory reading, independent study, and practical exercises.

Participants completing the EASSS 2026 Summer School will receive a Certificate of Attendance indicating a recommendation of 3 ECTS. The recommendation reflects participation in the summer school and its associated learning activities, including tutorial sessions, recommended readings, preparation materials, and exercises.

Recognition of the recommended ECTS is subject to the rules and approval procedures of the participant's home university or PhD programme.